

Task 1.1

Describe this transimpedance-amplifier circuit – how does it work? What is the task of every single component?

Transimpedance amplifier is a negative feedback amplifier which create a low input impedance at the input and suited for sensing current generated by sensors.

Transimpedance-amplifier also known as Current controlled Voltage source.

Transimpedance is can be used with Photodiode to detect the Light falls on it, by sensing the current generated by Photodiode and then amplify the signal.

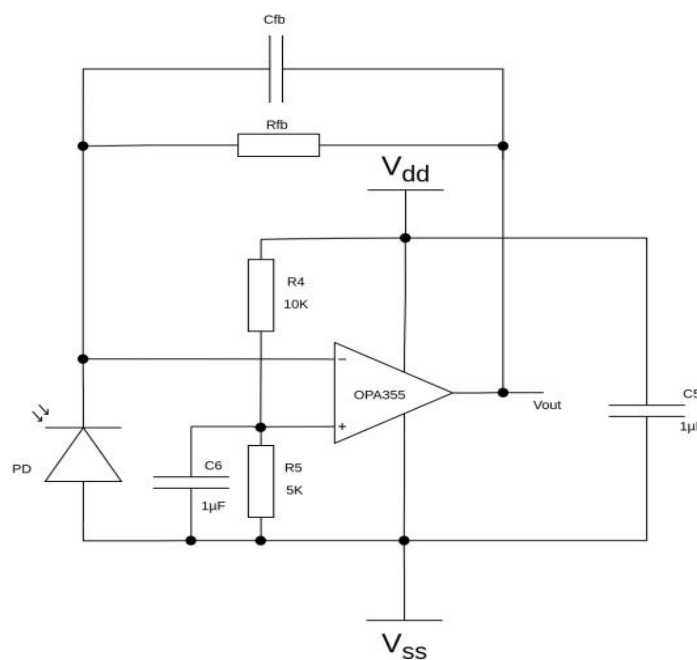


Fig01: Task 1.1 TIA Circuit with Photodiode.

PD (Photo Diode):

Photodiode is for detection of the light. An Optical Receiver needs a photodiode to convert the information carried by the modulated light to current. The diode suffers from a direct trade-off between its output current and parasitic capacitance (because the light increased by the diode is proportional to its area), which needs a virtual ground termination.

R_{fb} (feedback resistance):

It employs negative feedback to create a low input impedance. The circuit is suited to sensing current thus acting as a current to voltage converter. This circuit structure also introduces virtual ground which is useful in two cases:

- i. grounds the parasitic capacitance to have better bandwidth.
- ii. Reduces the voltage swings by a factor of $(1+A_o)$, where A_o = Open loop gain

Cfb (feedback capacitance):

Capacitance added to make an active low pass filter. When a frequency depended, element is added to the feedback loop of an inverting amplifier in a manner that the negative feedback increases for higher frequency, thus it works as a low pass filter.

R4, R5, C5, C6 (Bias Resistance and Capacitance):

By adding resistance and capacitance at the non-inverting input, performance and stability of TIA can be improved.

Some main purposes:

- i. Stability and Bandwidth control: addition of R&C in parallel of non-inverting input stabilizes OpAmp by compensating phase shift introduced by feedback. The RC network set required Bandwidth of the amplifier over required frequency range.
- ii. Noise Reduction: Resistor at non-inverting input helps to reduce noise gain. By carefully choosing the value of the resistor, its possible to balance the noise performance. A low pass filter configuration with amplifier input capacitance, reducing the high frequency noise can also be produced.
- iii. Prevention of saturation: To amplify properly Opamp must not go to saturation. Bias resistors prevent Opamp from being saturation region.
- iv. Improved impedance matching: resistance of non-inverting input required for impedance matching. Balancing of the impedance is important for reducing offset voltage by input bias current.
- v. Prevents high frequency oscillation and improves phase margin.
- vi. Limits current going into OpAmp and thus protects against damage due to high input currents.

Task 1.2

Simulate the circuit with different (dynamically changing) input currents from the photodiode.

Simulations are done in LTSpice. The Library for the OPA355 is downloaded from Texas Instruments. But library for photodiode *OSRAM LW VH8G* is not found yet. Although there is dataset to be found from OSRAM webpage, I couldn't understand and make library out of that. Instead I have used another Optodiodes which were available with the LTSPice Library, named MOC205. Source voltage is chosen as a Sinewave having, offset voltage = 0V, Amplitude = 2V, Frequency = 80k, Timedelay = 0, AC amplitude = 2V. The optodiode will transform the voltage source as a current source. The rest of the circuit values are chosen as the given Task.

The simulations files are added to with email.

Transient Analysis:

Here a non-linear, time-domain simulation is done known as Transient Analysis. Stop time is chosen as .1ms:

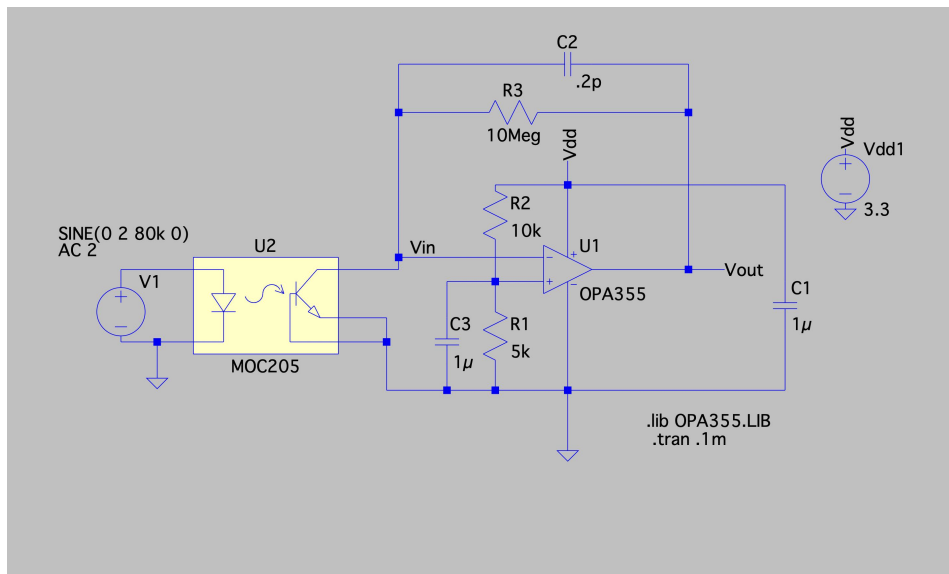


Fig02: Task 1.2 TIA Circuit for Transient Analysis

Comparison of Vin and Vout in Time domain:

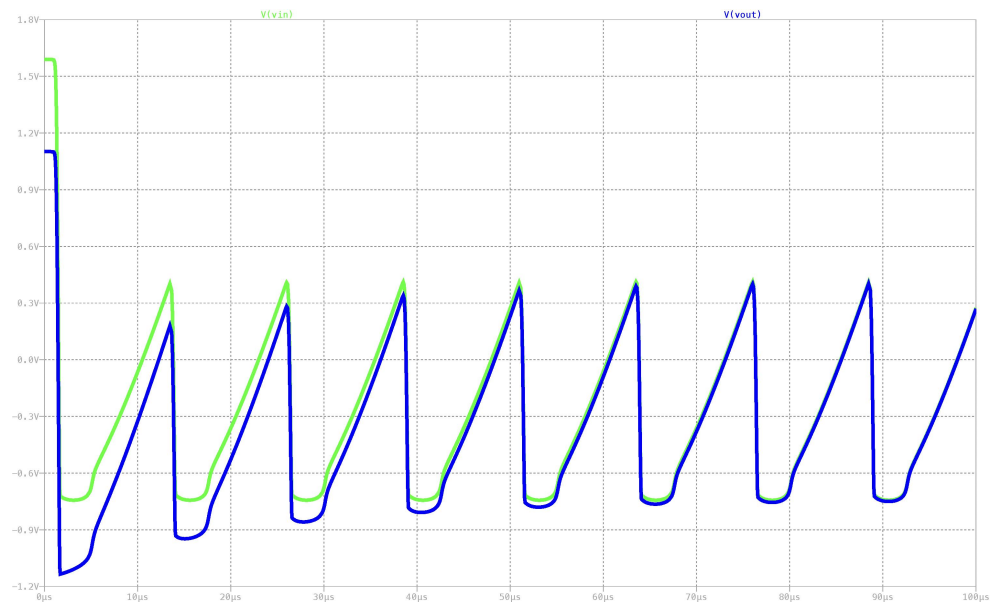


Fig03: Transient Analysis plot for TIA Circuit.

AC Analysis/Frequency Response:

Here the small signal AC behaviour/ Frequency response of the circuit is simulated. Nature of the sweep is in decade, number of points per decade is 1000, Start Frequency is 1Hz and Stop Frequency is 1MHz.

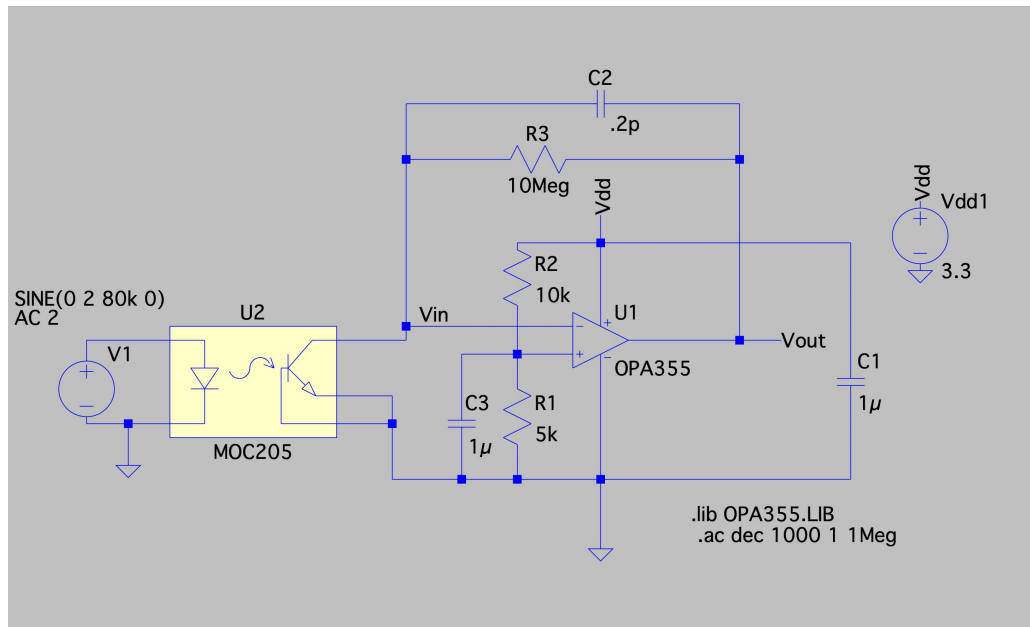


Fig04: Task 1.2 TIA Circuit for AC Analysis

Frequency response of Vin and Vout is compared both in Magnitude/Bandwidth and Phase Plot:

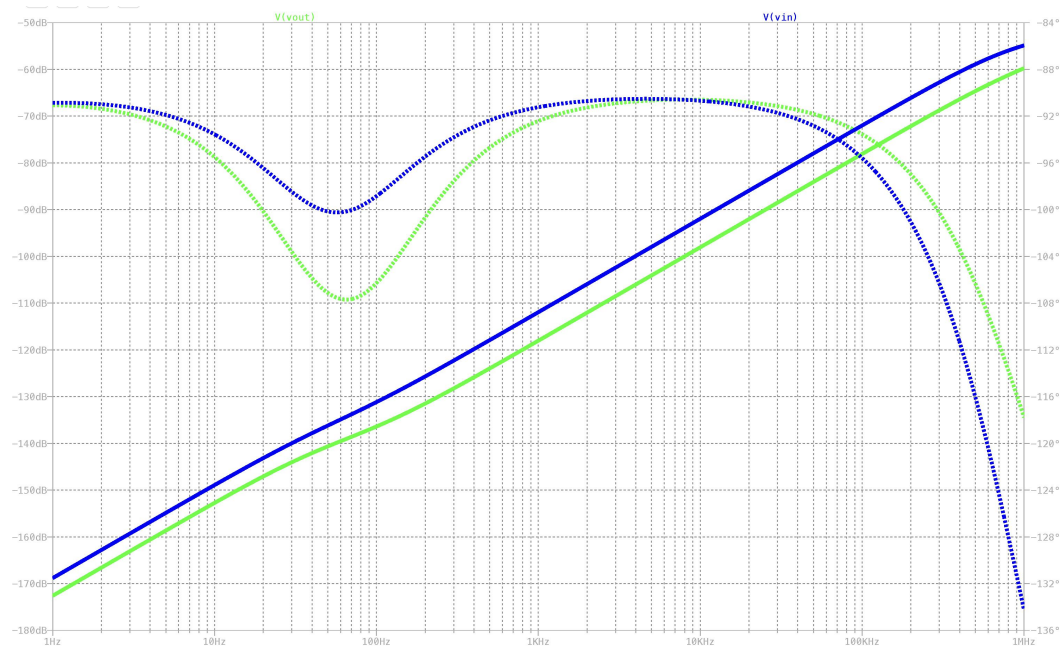


Fig05: AC Analysis plot for TIA Circuit.

